

Leo Shaposhnik



Overview

I am a theoretical physicist who builds agentic systems for open-ended scientific discovery. In this capacity, I design and build adversarial AI systems for the development of mathematical theories of physical systems and develop code for the large-scale simulation of many-body systems on HPC clusters. I have 10+ years of experience writing code, including 6+ years of experience contributing to academic codebases and 1+ year contributing to production codebases in industry as an IT consultant. I lead several research projects from initial conception through completion and publication in international journals. Over the past year I have developed a deep interest in knowledge representation, formal verification and mechanistic reasoning as approached by transformers, dependent type theory and category theory.

Current Position

PhD in Theoretical Physics

Berlin, Germany

Freie Universität Berlin

04/2023 – present

- Conduct research at the intersection of quantum information theory, tensor networks and high-energy physics.
- Experiment with multi-agent systems and context engineering for reliable collaboration on open-ended discovery.
- Implement parallelized high-performance code for the simulation of many-body systems on HPC clusters.
- Lead multiple international collaborations from the initial idea towards publication.
- Optimize RNNs and multiscale CNNs for the numerical study of quantum states.
- Co-organize a weekly seminar on quantum many-body physics.
- Teach courses on quantum mechanics, gravity and quantum information.

Previous Experience

IT Consultant

Berlin, Germany

d-fine GmbH

11/2021 – 03/2023

- Owned a core software component of a production-grade IFRS 17 reporting backend for an international reinsurer.
- Coordinated feature planning and implementation with client stakeholders.
- Supported testing, deployment and bug-fixing across the whole project.

Working Student — Mathematical Modeling and Data Analysis

Berlin, Germany

Physikalisch-Technische Bundesanstalt (PTB)

07/2020 – 10/2021

- Formulated theoretical models to study roundoff-induced effects in classical chaotic systems.
- Performed high-precision long-time simulations of chaotic systems on an HPC cluster.

Working Student

Zeuthen, Germany

DESY

03/2017 – 06/2020

- Designed and set up optical experiments to measure radiation damage in optical components for a linear accelerator.
- Analyzed the concentration of radioactive material in accelerator components with MATLAB.

Education

Master's Degree — Physics (Avg. Grade: 1.0 / GPA: 4.0)

Berlin, Germany

Humboldt-Universität zu Berlin / Albert Einstein Institute

10/2019 – 02/2022

Thesis on entanglement entropy in curved spacetime, including analytical derivations and numerical validation using tensor network techniques.

Bachelor's Degree — Physics (Avg. Grade: 1.7 / GPA: 3.3)

Berlin, Germany

Humboldt-Universität zu Berlin

10/2016 – 09/2019

Thesis on Hopf-algebraic structures underlying perturbative renormalization in quantum field theory.

Technologies

- **Python:** PyTorch, Triton, SciPy, Hugging Face Transformers
- **Agent systems:** OpenAI API, Anthropic API, LangGraph, MCP, RAG
- **Scientific computing:** Julia, MATLAB, Wolfram Language
- **High-performance computing:** C++, CUDA, Linux, Slurm
- **Data:** SQL, SAS, Pandas
- **Version control & CI/CD:** Git, Github, Azure DevOps
- **Formal methods:** Lean 4

Awards & Honors

- Honorable mention for my poster “On Infinite Tensor Networks, Complementary Recovery and Type II Factors”. QIQG 2025, Perimeter Institute.

Publications

- **On Infinite Tensor Networks, Complementary Recovery and Type II Factors.** W. Chemissany, E. Gesteau, A. Jahn, D. Murphy, **L. Shaposhnik**. J. Phys. A 58 (2025) 435301.
- **Searching for Emergent Spacetime in Spin Glasses.** D. Saraidaris, **L. Shaposhnik**. JHEP 04 (2026) 032.

Languages

- **English** (Fluent)
- **German** (Native)
- **Russian** (Native)
- **Ukrainian** (Beginner)